

Sliding Mode Observer-Based Mechanical Parameter Estimation for Permanent-Magnet Synchronous Motor

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- **Abstract**—A mechanical parameter estimation algorithm for permanent magnet synchronous motor (PMSM) drive systems basing on sliding mode observer is proposed in this paper. First, an extended sliding mode mechanical parameter observer (ESMMPO) is presented to track system disturbances in real time which include the information of mechanical parameters. Based on this ESMMPO, the mechanical parameter can be extracted from estimated system disturbance. Therefore a simple algorithm for mechanical parameter estimation of the PMSM control system is presented. The parameter choice guidelines of ESMMPO are designed to guarantee the global stability of the observer. Then, the equivalent low-pass filter function of the ESMMPO is presented to suppress sliding mode chattering, which allows softening output signal of observer while maintain no phase lag incurring. Therefore, the smooth output of the ESMMPO can be directly used for the parameter estimation. Experimental results show the validity of the proposed parameter estimation approach.

Index Terms—Parameter estimation, extended state, permanent-magnet synchronous motor (PMSM), sliding mode observer.

I. INTRODUCTION

Permanent magnet synchronous machines (PMSMs) are now widely used in many applications, ranging from electric vehicle, industrial drives, and wind power generation to aerospace, due to their high precision, high efficiency, high speed, high reliability and excellent control performance. In a high-performance control system of PMSM, parameter accuracy is an important factor that influences control precision of the whole system. Specially, mechanical parameters, including the moment of inertia of the whole system, the viscous damping coefficient, and the load torque, play a key role in achieving high dynamic performance of the speed control. If these parameters can be estimated accurately, the autotuned controller of PMSM systems can be achieved easily, which means optimal control method can be designed to improve the control performance of the PMSM systems.

Therefore, many parameter estimation methods have been proposed, and basically can be categorized into five groups: parameter adaptation [1]-[5], model reference adaptive system (MRAS) [6][11], Recursive Least Squares (RLS)[7][8], observer based methods and extended Kalman filter (EKF) [9]-[13]. The MRAS-based method is simple and easy to implement, which has been used for estimating parameter in many applications[14]-[18]. However, this method cannot be applied for estimating the load torque in real time. Parameter adaptation method can be used for estimating parameter by adjusting adaptive law or function cost continuously. However, it is difficult for real application because of the complexity in practical implementation and the sensitivity of adaptation gains. EKF algorithm treats mechanical parameter as

one of the system variables, and such estimates are obtained as a direct output of EKF, while RLS iteratively solves the Least Squares estimates upon convergence which relatively much depends on initial conditions. Although RLS method can be applied for estimating parameter of a PMSM control system, it requires a long estimated time and a special trajectory of reference velocity during the estimation. Therefore, this method is limited in some practical applications [4], [19].

Observer based approaches, including disturbance observer (DOB) and sliding mode observer (SMO), are attracting widespread interest, because its design and implementation is relatively simple for parameter estimation. The DOB method can be applied for estimating viscous friction coefficient, rotational inertia and load disturbances by means of building disturbance estimators [20]-[23]. In [20], a state and load torque observer is developed to observe the unknown load torque in real time. In [23], a state observer for estimating motor disturbance and mechanical parameters is presented. In this method, two adaptive controllers following the state estimator are used to separately adjust the drive inertia and friction to their correct value. Although the DOB methods in [20] and [23] can achieve good performance and are able to estimate the mechanical parameters accurately, both of them have poor robustness. Compared to DOB method, SMO has attractive advantages of robustness to disturbances and low sensitivity to the parameter variations of the whole system. Therefore, many SMO based parameter estimation methods have been proposed. In [24], a novel SMO is designed to estimate the friction state of the LuGer model. In [25] and [26], in order to improve estimation accuracy, the second-order SMO is

designed for the identification of mechanical systems. However, in aforementioned observers, the low-pass filter must be used to obtain estimated parameters, which will lead to the phase lag and the amplitude attenuation, then reduce the estimation accuracy.

In this paper, an extended sliding mode observer-based method for estimating the mechanical parameters of the PMSM drive system is proposed. First, the system disturbances, which include parameter error of viscous friction coefficient, parameter error of rotational inertia and load disturbances, are regarded as extended system state to build extended state equation, which is used to design the extended sliding-mode mechanical parameter observer (ESMMPO). Based on the proposed ESMMPO, a simple algorithm for mechanical parameter estimation of the PMSM control system is developed. This ESMMPO can track system states in real time, and can be equivalent to a low pass filter, which can soften switching control signal of sliding mode. Therefore, the output of the ESMMPO can be directly used for the mechanical parameter estimation with no phase lag incurring. Finally, the effectiveness of the proposed parameter estimation approach was verified by experimental results.

II. MACHINE MODEL

Assuming that the employed SPMSM has negligible crosscoupling magnetic saturation, structural asymmetry, iron losses, magnet eddy current loss, and harmonics in the descriptive functions of windings, rotor anisotropy, and coercive force of magnets, the dynamics of a SPMSM in the rotor d - q coordinates can be described as follows [27]:

$$\begin{cases} u_d = Ri_d - p\omega Li_q + L\dot{i}_d \\ u_q = Ri_q + p\omega Li_d + p\omega\psi_a + L\dot{i}_q \end{cases} \quad (1)$$

$$T_e = 1.5p\psi_a i_q. \quad (2)$$

$$T_e - T_L = J\dot{\omega} + B\omega. \quad (3)$$

Where u_d and u_q represent d- and q-axis stator voltages; i_d and i_q are d- and q-axis currents; L is stator inductance; R is stator winding resistance; T_e is electrical magnetic torque ($N\cdot m$); T_L is load torque ($N\cdot m$); p is number of pole pairs; ψ_a is flux linkage of permanent magnets; ω is angular velocity (rad/s); B is viscous friction coefficient; J is rotational inertia ($kg\cdot m^2$).

Typical structure diagram of PMSM vector control system is shown as Fig. 1. The power part is comprised of the PWM voltage source inverter and PMSM. The stator currents and motor speed can be directly measured by current sensors and position encoder, respectively. The control parts employ a structure of cascade control loops including a speed loop and two current loops, i.e., speed outer loop and two current inner loops. The current inner loop is comprised of d-axis control loop and q-axis control loop, which can be used to control d-axis and q-axis current (i_d and i_q), respectively. The currents i_d and i_q can be calculated from i_a and i_b (which can be measured by current sensors) by Clarke and Park transformations. Usually, the reference current i_d^* and i_q^* are set to be zero and output of speed loop, respectively. Moreover, the outputs of d-axis and q-axis current loop are used to generate drive signal of inverter by Park inverse transformations and space vector pulse width modulation (SVPWM).

Generally, the PI controller in speed loop and the PI controller in current loop, as

shown as Fig.1, are designed based on equations (1) and (3), which represent PMSM electrical model and mechanical model, respectively. Therefore, the parameter accuracy of both the electrical and mechanical models is very important for controller design in a PMSM system. The electrical model (1) includes parameter R , L , p and ψ_a , which can be expressed as electrical parameters. The mechanical model (3) includes parameter B , J and T_L , which can be expressed as mechanical parameters. The mechanical parameters of a PMSM system are unmeasurable but play a key role in controller design of speed loop, which will determine both the dynamical and steady-state performances of a PMSM system. However, compared with the mechanical parameters, the electrical parameters can be easily measured using instruments or sensors. Therefore, this paper focuses on the estimation of the mechanical parameters J , B , and T_L .

III. EXTENDED SLIDING MODE MECHANICAL PARAMETER OBSERVER

According to the equation (3), the dynamic equation of the motor can be expressed as follows, with the system disturbances taken into account:

$$J_0 \dot{\omega} = T_e - B_0 \omega + d \quad (4)$$

where $J = J_0 + \Delta J$ and $B = B_0 + \Delta B$. B_0 and J_0 are crude estimations of the true parameters, which can be determined initially based on the experiences and prior knowledge. ΔJ and ΔB are parameter errors between the true system and their crude estimations.

In equation (4), d represents the disturbances including parameter errors and load disturbances, i.e.,

$$d = -\Delta J \dot{\omega} - \Delta B \omega - T_L \quad (5)$$

Furthermore, disturbances d are regarded as extended system state, and then dynamic model (4) can be rewritten as the following extended state equation:

$$\begin{cases} J_0 \dot{\omega} = T_e - B_0 \omega + d \\ \dot{d} = r \end{cases} \quad (6)$$

where r is the variation rate of system disturbances d . According to equation (5), r can be described as $r = -\Delta J \ddot{\omega} - \Delta B \dot{\omega} - \dot{T}_L$.

For the purpose of parameter estimation, the ESMMPPO (extend sliding-mode mechanical parameter observer) can be designed as below:

$$\begin{cases} J_0 \dot{\hat{\omega}} = T_e - B_0 \hat{\omega} + \hat{d} + u_{smo} \\ \dot{\hat{d}} = m u_{smo} \end{cases} \quad (7)$$

where $\hat{d}$ is an estimate of disturbances d , $\hat{\omega}$ is an estimate of speed ω , m is sliding-mode parameter which is discussed in next section, and u_{smo} represents sliding mode switching signal that is designed as

$$u_{smo} = \eta \cdot \text{sgn}(S) \quad (8)$$

where η is negative and S is sliding-mode surface, which is designed as $S = \hat{\omega} - \omega$.

From (6) and (7), the error equation can be obtained as

$$\begin{cases} J_0 \dot{\hat{\omega}} - J_0 \dot{\omega} = B_0 \omega - B_0 \hat{\omega} + \hat{d} - d + u_{smo} \\ \dot{\hat{d}} - \dot{d} = m u_{smo} - r \end{cases} \quad (9)$$

Define $e_1 = \hat{\omega} - \omega$ and $e_2 = \hat{d} - d$. Then equation (9) can be rewritten as

$$\begin{cases} J_0 \dot{e}_1 = -B_0 e_1 + e_2 + u_{smo} \\ \dot{e}_2 = m u_{smo} - r \end{cases} \quad (10)$$

Based on the above descriptions and discussions, the parameter choice guidelines of ESMMPPO and sliding-mode chattering suppression analysis are introduced as

follows.

A. Choice of Observer Parameter

In order to ensure sliding mode occurring, parameter η of observer should be selected reasonably, which means the stability condition of sliding-mode observer must be satisfied. Therefore, the following Lyapunov function is considered: $V=0.5s^2$.

Differentiating V with respect to time t gives

$$\dot{V} = s \cdot \dot{s} = e_1 \cdot \dot{e}_1. \quad (11)$$

From (10), equation (11) can be rewritten as

$$\begin{aligned} \dot{V} &= \frac{1}{J_0} e_1 (-B_0 e_1 + e_2 + u_{smo}) \\ &= \frac{1}{J_0} e_1 [(e_2 - B_0 e_1) + \eta \cdot \text{sgn}(e_1)] \\ &= \begin{cases} \frac{1}{J_0} e_1 [(e_2 - B_0 e_1) + \eta] & e_1 > 0 \\ \frac{1}{J_0} e_1 [(e_2 - B_0 e_1) - \eta] & e_1 < 0 \end{cases} \end{aligned} \quad (12)$$

To ensure the stability of ESMMPPO, stability condition $\dot{V} = s \cdot \dot{s} = e_1 \cdot \dot{e}_1 < 0$ must be satisfied, then (12) can be expressed as

$$\begin{aligned} \dot{V} &= \begin{cases} \frac{1}{J_0} e_1 [(e_2 - B_0 e_1) + \eta] & e_1 > 0 \\ \frac{1}{J_0} e_1 [(e_2 - B_0 e_1) - \eta] & e_1 < 0 \end{cases} \\ &< 0 \end{aligned} \quad (13)$$

As a result,

$$\eta < -|e_2 - B_0 e_1|. \quad (14)$$

Relation (13) and (14) show that, in order to ensure observer stability, the selection of parameter η need to meet limiting condition. In practical application, the following

parameter adaptive law can be used:

$$\eta = -l|e_2 - B_0 e_1|, \quad l > 1 \quad (15)$$

where l is safety factor of the sliding mode. Usually, $l=2$ is enough to ensure the observer stability.

It can be seen that, error e_1 and its derivative $\dot{e}_1$ can converge to zero along sliding mode occurring in finite time, which means $e_1 = \dot{e}_1 = 0$. Therefore, error equation (10) can be simplified as

$$\begin{cases} e_2 = -u_{smo} \\ \dot{e}_2 = mu_{smo} - r \end{cases} \quad (16)$$

which can also be rewritten as

$$\dot{e}_2 + me_2 + r = 0. \quad (17)$$

Thus, the solution for e_2 can be obtained as bellow:

$$e_2 = e^{-mt} [C + \int r \cdot e^{mt} dt] \quad (18)$$

where C is a constant. To ensure the disturbance estimation error e_2 converge to zero, the sliding-mode parameter is selected as

$$m > 0. \quad (19)$$

Moreover, the convergence rate of error e_2 is in direct relation with parameter m .

In the above analysis, general ideas about the role of ESMMPPO parameters and the way they can be chosen in the observer design are discussed. It can be seen that the choice of the ESMMPPO parameters is only constrained by relations (14) and (19). The schematic block diagram of the ESMMPPO is shown as Fig. 2.

B. Equivalent Low-Pass Filter Analysis

Speed can be described as $\omega = \hat{\omega}$ after sliding mode occurring. Then, equation (7) can be simplified as

$$\begin{cases} J_0 \dot{\omega} = T_e - B_0 \omega + \hat{d} + u_{smo} \\ \dot{\hat{d}} = m u_{smo} \end{cases} \quad (20)$$

Then, based on equation (6), the following equation can be obtained:

$$\dot{\hat{d}} + m\hat{d} = md \quad (21)$$

Equation (21) is equivalent to the LPF (low-pass filter), whose transfer functions $F(s)$ are given as follows:

$$\hat{d} = \frac{m}{s + m} \cdot d \quad (22)$$

It can be seen that disturbance observation effect of ESMMPPO is equivalent to the output of filtering real system disturbance. The cutoff frequency of LPF is m , which can be designed arbitrarily based on the requirement of chattering suppression in the observer, since the condition (19) are satisfied (the effect of cut-off frequency on the performance of parameter estimation is discussed in Section V.). Thus, output of the proposed observer doesn't include the sliding-mode chattering because of the LPF and can be directly used for the system control.

According to above analysis, proposed ESMMPPO can be equivalent to a LPF after sliding mode occurring, then, the schematic block diagram of the ESMMPPO shown in Fig.2 (a) can be simplified as Fig. 2(b), which shows the schematic block diagram of equivalent LPF.

IV. PARAMETER ESTIMATION SCHEME

According to above analysis, it can be find that the proposed ESMMPPO can observe real-time information of system disturbances, which include mechanical parameters of PMSM control system, at the same time, inherent chattering phenomenon of sliding-mode can be suppress effectively. Therefore, the mechanical parameter can be extracted from estimated system disturbance.

The estimation principle is shown in Fig. 3 and can be described as follows: Firstly,

at the stage of estimating B , as shown in Fig. 3(a), PMSM should be operated at two different steady velocities, and only measured velocity information and estimated disturbances are used for estimating parameter B during this stage; Secondly, at the stage of estimating J , as shown in Fig. 3(b), PMSM should be operated at two different constant accelerations or decelerations, respectively, and only accelerations or decelerations and estimated disturbances are used for estimating parameter J during this stage; Finally, after estimating B and J , estimated parameters are required to update to equation (5), which can be used to estimate load torque T_L directly.

However, it should be noted that before parameter estimation, optimal PI speed controller cannot be obtained because exact parameters B , J and T_L are unknown. Therefore, trial-and-error method based nonoptimal PI speed controller can be used for mechanical parameter estimation. Obviously, good performance of the PMSM system cannot be obtained using this speed controller. However, since the parameter estimation method needs only the steady state information of the system, the nonoptimal PI controller has little effect on the parameter estimation.

Based on ESMMPPO, equation (4) and (5), the following method is proposed for mechanical parameters estimation in the PMSM system. The structure diagram of parameter estimation is shown as Fig. 4.

Step 1: Estimation of Parameter B

According to equation (4) and (5), estimated parameter B can be expressed as $\hat{B} = B_0 + \Delta\hat{B}$. Therefore, one can estimate $\hat{B}$ by estimating parameter error $\Delta\hat{B}$, because crude estimation of parameter B_0 is known. When estimating B , as shown in

Fig.3 (a), PMSM is required to operate at two different steady speeds, which satisfying $\omega(t) \neq \omega(t + \tau)$ and $\dot{\omega}(t) = \dot{\omega}(t + \tau) = 0$. In Fig.3, τ is a constant time delay, which means that two different steady speeds correspond to two different periods of time, respectively. The value of τ is 1 second, and τ has no effect on the performance of parameter estimation.

When the motor operating at the first steady speed $\omega(t)$, based on ESMMPPO and equation (5), the disturbance estimation can be obtained as bellow:

$$\hat{d}(t) = -\Delta\hat{J}\dot{\omega}(t) - \Delta\hat{B}\omega(t) - T_L. \quad (23)$$

Then, delaying (23) by a time delay τ , the disturbance estimation for the second steady speed $\omega(t + \tau)$ is given as follows:

$$\hat{d}(t + \tau) = -\Delta\hat{J}\dot{\omega}(t + \tau) - \Delta\hat{B}\omega(t + \tau) - T_L. \quad (24)$$

When the PMSM control system reaches its steady state speed, load torque T_L can be considered as a constant. Then, subtracting (23) from (24) gives

$$\hat{d}(t + \tau) - \hat{d}(t) = -\Delta\hat{B}[\omega(t + \tau) - \omega(t)]. \quad (25)$$

Thus, $\Delta\hat{B}$ can be obtained as below:

$$\Delta\hat{B} = -\frac{\hat{d}(t + \tau) - \hat{d}(t)}{\omega(t + \tau) - \omega(t)}. \quad (26)$$

Therefore, estimated parameter $\hat{B}$ can be obtained as follows:

$$\hat{B} = B_0 + \Delta\hat{B} = B_0 - \frac{\hat{d}(t + \tau) - \hat{d}(t)}{\omega(t + \tau) - \omega(t)}. \quad (27)$$

Step 2: Estimation of Parameter J

After estimating B , crude estimation B_0 in (7) can be updated by $\hat{B}$. Then according to (5) and (7), disturbance estimation $\hat{d}$ can be rewritten as

$$\hat{d} = -\Delta\hat{J}\dot{\omega} - T_L. \quad (28)$$

When estimating J , as shown in Fig.3 (b), the motor need to operate at the first constant acceleration state ac_1 . In this case, based on ESMMPPO and equation (5), the disturbance estimation can be obtained as bellow:

$$\hat{d}(t) = -\Delta\hat{J} \cdot ac_1 - T_L. \quad (29)$$

Then, delaying (29) by a time delay τ , which ensure two different constant accelerations correspond to two different periods of time, respectively, the disturbance estimation for the second constant acceleration state ac_2 is given as follows:

$$\hat{d}(t + \tau) = -\Delta\hat{J} \cdot ac_2 - T_L. \quad (30)$$

Subtracting (30) from (29) gives

$$\hat{d}(t + \tau) - \hat{d}(t) = -\Delta\hat{J}(ac_2 - ac_1). \quad (31)$$

Thus, $\Delta\hat{J}$ can be obtained as below:

$$\Delta\hat{J} = -\frac{\hat{d}(t + \tau) - \hat{d}(t)}{ac_2 - ac_1}. \quad (32)$$

Therefore, estimated parameter $\hat{J}$ can be obtained as follows:

$$\hat{J} = J_0 + \Delta\hat{J} = J_0 - \frac{\hat{d}(t + \tau) - \hat{d}(t)}{ac_2 - ac_1}. \quad (33)$$

In a practical PMSM control system, there exist load disturbances, which may vary as operation conditions vary. The control performance of a PMSM system will reduce seriously, if load disturbance cannot be suppressed effectively. Especially, it will be very difficult to limit load disturbances rapidly if adopting linear control algorithms like PI control method [27]. Therefore, in order to achieve excellent performance, it is very important for the PMSM control system to estimate load torque in real time.

Step 3: Estimation of Load Torque T_L

When the parameter B and J is estimated accurately, crude estimation B_0 and J_0 in (7) can be replaced by estimated parameters, i.e., $\Delta J = J - \hat{J} = 0$ and $\Delta B = B - \hat{B} = 0$. Then, the ESMMPPO (7) can be rewritten as

$$\begin{cases} \hat{J}\dot{\hat{\omega}} = T_e - \hat{B}\hat{\omega} + \hat{d} + u_{smo} \\ \dot{\hat{d}} = mu_{smo} \end{cases} \quad (34)$$

Where the updated disturbance estimation $\hat{d}$ can be expressed as

$$\hat{d} = -T_L. \quad (35)$$

From (34) and (35), it can be seen that the updated ESMMPPO have the ability of estimating load torque. Therefore, in practical application, the proposed ESMMPPO can be used for estimating B and J , and can also be used for online real-time estimating load torque T_L when parameter B and J are known.

According to aforementioned parameter estimation step, it can be observed that the proposed method is simple and easy to implement. Moreover, this method does not need bidirectional rotating operation, thus it can be very useful especially in the control system such that only the unidirectional rotation is allowed. The structure diagram of this parameter estimation method is shown in Fig.4.

V. SIMULATION AND EXPERIMENTAL RESULTS

In this section, to evaluate the performance of the proposed parameter estimation approach, the simulation and experimental system for speed control of PMSM was built. Simulations are established in MATLAB/Simulink. The experiments platform is constructed by TMS320LF2812 processor and the system sampling frequency is 10kHz. The photograph of the experimental setup including the PMSM, the motor

drive, the load motor and the control PC are shown in Fig. 5. In addition, the parameters of PMSM used in the experiment are listed in table I.

The simulation results of the estimated B with different speed conditions are shown in Figs. 6 and 7. From the simulation results, it can be observed that the proposed method can estimate parameter B exactly when different speed conditions are utilized. Moreover, the simulation results of estimated J with different accelerations are shown in Fig. 8 and 9. When acceleration state is positive, the estimated parameter J can converge to the true value, on the other hand, the similar results also can be obtained when acceleration state is negative. In addition, estimated load disturbances are shown in Fig. 10, when load torque T_L is increased suddenly from 1.2 N·m to 3N·m and decreased from 3N·m to 1.2 N·m, respectively. It can be observed that the observer can estimate the load disturbance exactly and quickly with low chattering.

The experimental measurements of the motor speed, the driving torque and the estimated disturbance are shown in Fig. 11 when the parameter B is estimated. Fig. 12 shows the experimental measurements of the motor speed, the driving torque and the estimated disturbance when the parameter J is estimated. The mechanical parameter estimation results are shown in Figs. 13-15. The estimation of B is displayed in Fig. 13. There are four different crude estimations of the true parameter of B , i.e., $B_0=0.0001B$, $B_0=0.01B$, $B_0=1.33B$ and $B_0=50B$, corresponding to Fig. 13(a)-(d). It is obvious that all the estimation curves of parameter B converge to the true parameter value. Similarly, Fig. 14 show the estimations of J with four different crude estimations, i.e., $J_0=0.001J$, $J_0=0.01J$, $J_0=2.5J$ and $J_0=10J$, corresponding to Fig.

14(a)-(d). It can also be found that all the estimation curves of parameter J can converge to the true value. Moreover, the estimation of load torque is shown in Fig. 15(a) and (b) when the load torques $T_L=1.2\text{Nm}$ and $T_L=3\text{Nm}$ are added respectively. Fig. 15(c) shows the dynamic response of estimated torque, when the load torque T_L increases from $1.2\text{N}\cdot\text{m}$ to $3\text{N}\cdot\text{m}$ suddenly. On the other hand, when the load torque T_L decreases from $3\text{N}\cdot\text{m}$ to $1.2\text{N}\cdot\text{m}$ suddenly, the dynamic response of estimated torque is shown in Figs. 15(d). It should be noted that proposed method can estimate load torque exactly and quickly, which means that estimated load torque can be used to develop disturbance rejection technique in a PMSM control system

In order to evaluate the performance of the proposed estimator at zero speed and during speed reversals, the experiment results with different speed conditions are shown in Fig.16, Fig.17 and Fig.18. There are two different crude estimations of the true parameter of B and J , respectively, i.e., $B_0=0.0001B$, $B_0=50B$, $J_0=0.001J$ and $J_0=10J$, corresponding to Fig. 16(c), Fig. 16(d), Fig. 17(c) and Fig.17 (d). The estimations of B with zero speed are displayed in Fig. 18. Fig.18 (a) and (b) show the speed responses and estimated B when speed change from 0rpm to 1200rpm . Moreover, when speed change from 0rpm change to -1200rpm , the speed responses and estimated B are shown in Fig.18 (c) and (d). According to experiment results, it can be found that the proposed estimation technique perform well at zero speed and during speed reversals.

The experiment results of parameter estimation with three different cut-off frequency of equivalent LPF, i.e., $m=2$, $m=5$ and $m=10$, are shown in Fig.19. From the

results, it can be seen that the lower cut-off frequency indicates higher performance of parameter estimation. Then, one can adjust the cutoff frequency of equivalent LPF based on the requirement of chattering suppression in the observer, since the condition (19) are satisfied.

In order to prove the occurrence of sliding mode, the experiment results using estimated speed as speed feedback are displayed in Fig.20, which shows the estimated speed and measured speed waveforms when speed feedback signal switching from measured speed to estimated speed at 1 second. In addition, PI algorithm is designed as the speed controller, which can be expressed as $G_s = \frac{k_p \cdot s + k_i}{s}$. In this speed PI

controller, $k_p = \frac{J \cdot \omega_{sc}^*}{k_t}$ and $k_i = \frac{B \cdot \omega_{sc}^*}{k_t}$ represent proportional gain and integral gain, respectively; $k_t = \frac{3}{2} p \psi_a$ represents torque constant and ω_{sc}^* is expected open-loop cutoff frequency of speed PI controller. Parameter J and B in gain k_p and k_i of speed PI controller can be adjusted using estimated mechanical parameters $\hat{J}$ and $\hat{B}$ based on the proposed observer. The experimental waveforms of utilizing estimated parameter in speed controller are shown in Fig.21, when load torque suddenly change from 2 N.m to 0 N.m.

From the experimental results, it is obvious that the estimated parameters can track true values exactly with different crude estimations, speed and load conditions, which verifies the effectiveness of the proposed estimation method.

VI. CONCLUSIONS

In this paper, one novel mechanical parameter estimation algorithm is proposed and

has been experimentally applied to a PMSM system. The major contributions of this work include: 1) in order to estimate parameters exactly, one extended sliding-mode mechanical parameter observer is presented; 2) the equivalent low-pass filter in the ESMMPPO can be used to control chattering, which means that the extra low-pass filter can be avoided and the output signal can be directly used for the parameter estimation; 3) parameter estimation step is designed, which does not need bidirectional rotating operation; and 4) the load torque can be estimated in real time. Experimental results have validated the proposed method.

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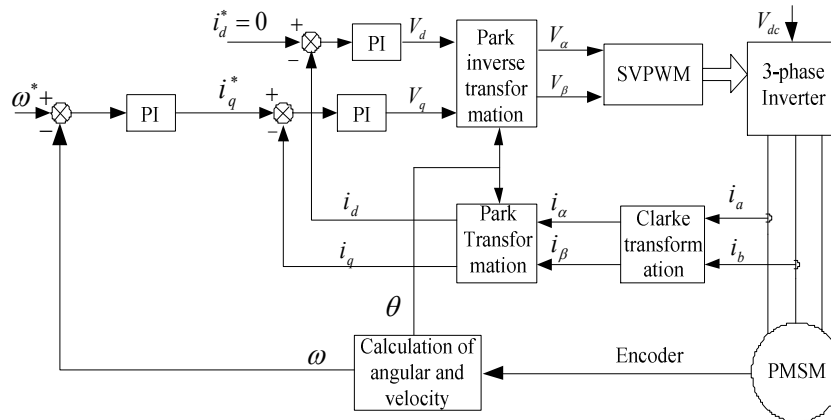
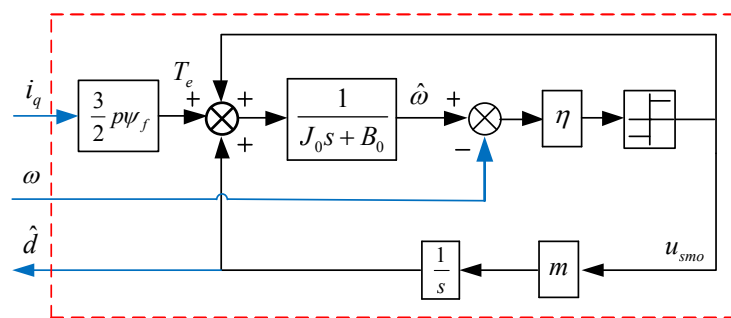


Fig.1. Structure diagram of the PMSM vector control system.



(a)

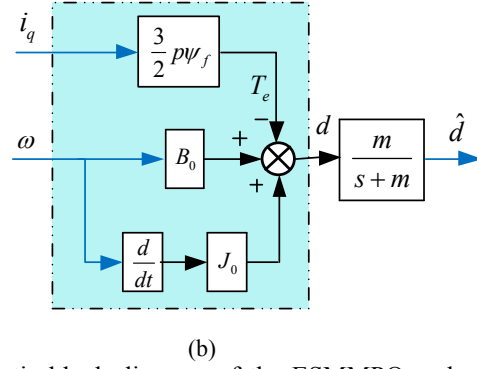


Fig. 2 Schematic block diagram of the ESMMPPO and equivalent low-pass filter. (a) ESMMPPO. (b) Equivalent low-pass filter after sliding mode occurring.

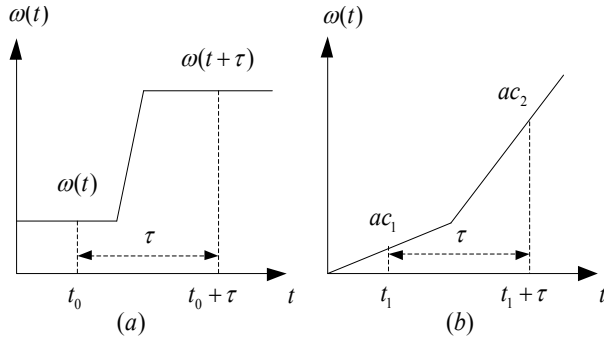


Fig. 3 Principle diagram of mechanical parameter estimation. (a) Estimating B . (b) Estimating J .

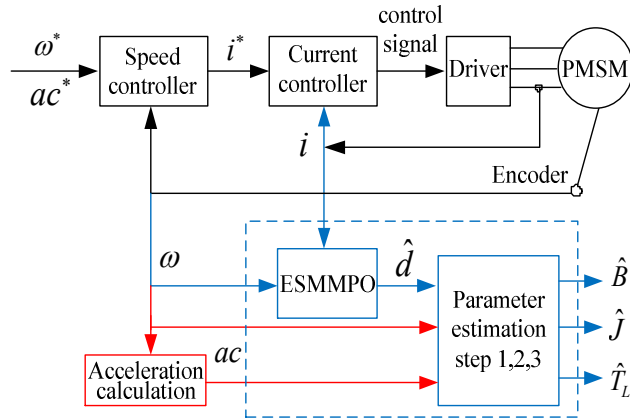


Fig.4. Structure diagram of the parameter estimation.



Fig. 5. Photograph of the experimental setup.

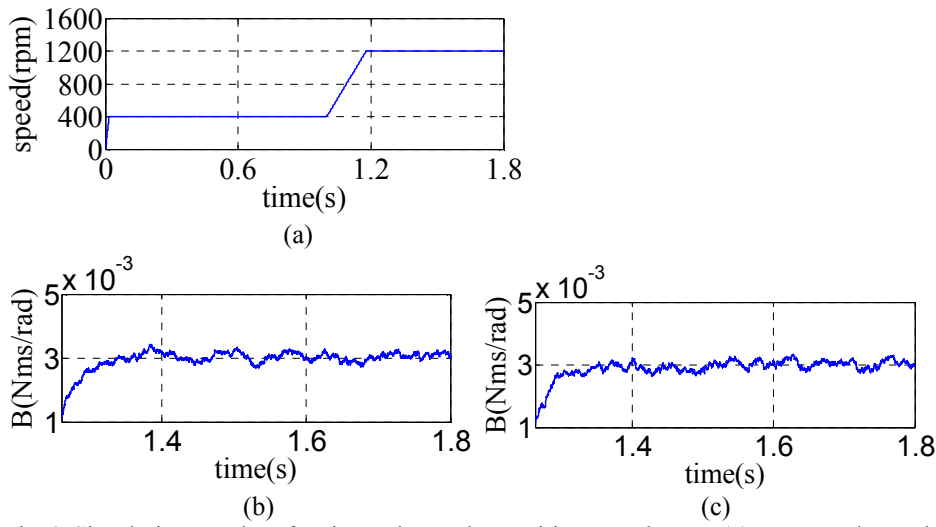


Fig. 6. Simulation results of estimated B under positive speed state. (a) Measured speed. (b) $B_0 = 0.0001B$. (c) $B_0 = 50B$.

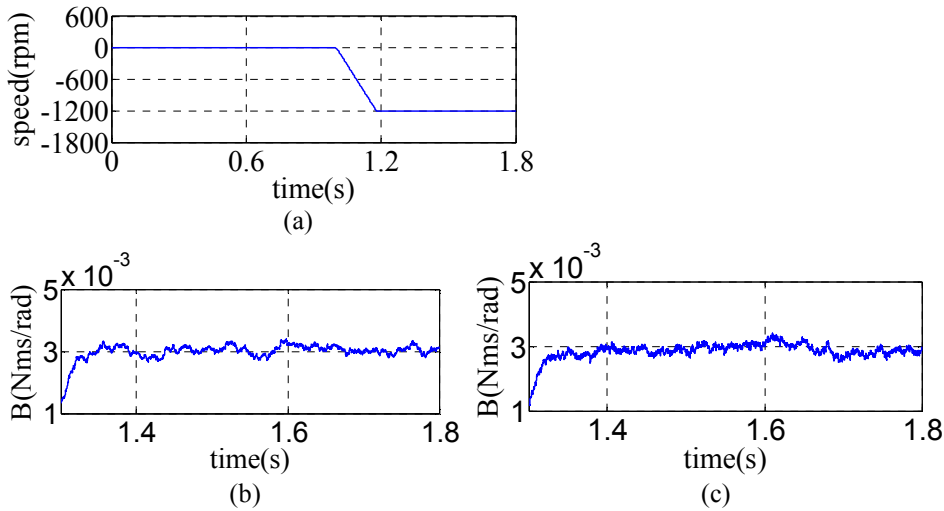


Fig. 7. Simulation results of estimated B under zero speed and speed reversals. (a) Measured speed. (b) $B_0 = 0.0001B$. (c) $B_0 = 50B$.

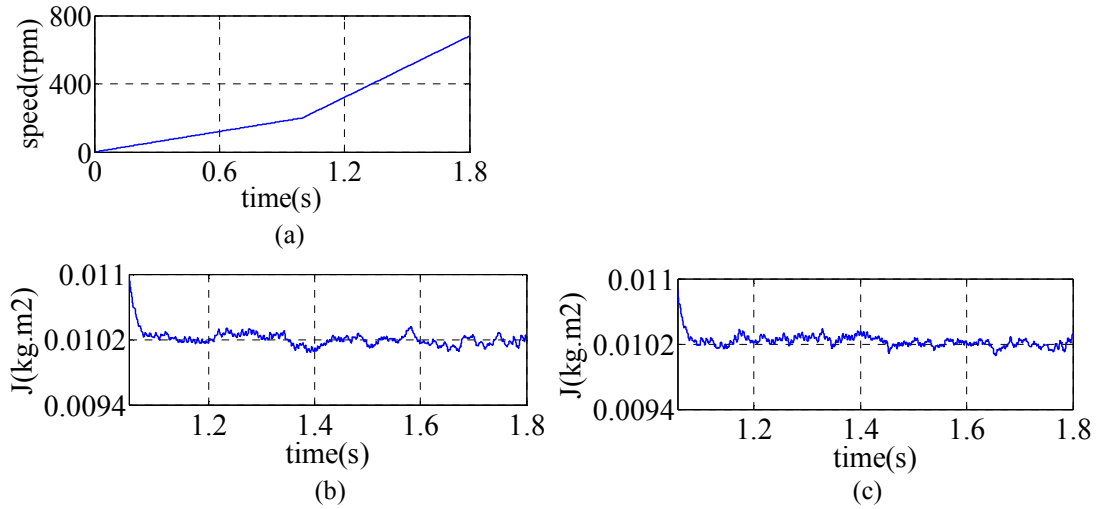


Fig.8. Simulation results of estimated J under positive acceleration state. (a) Measured speed. (b) $J_0=0.001J$. (c) $J_0=10J$.

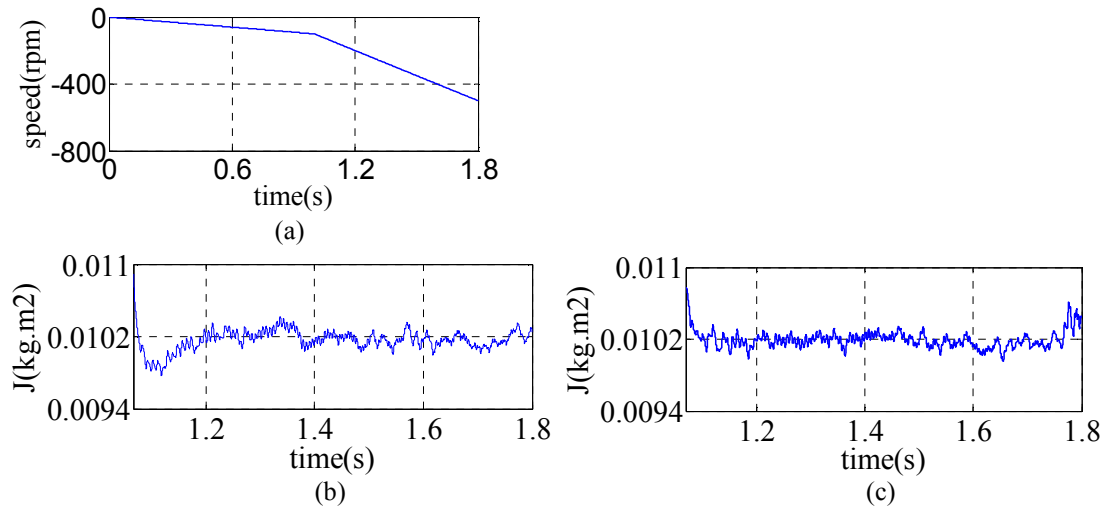


Fig.9. Simulation results of estimated J under acceleration reversals. (a) Measured speed. (b) $J_0=0.001J$. (c) $J_0=10J$.

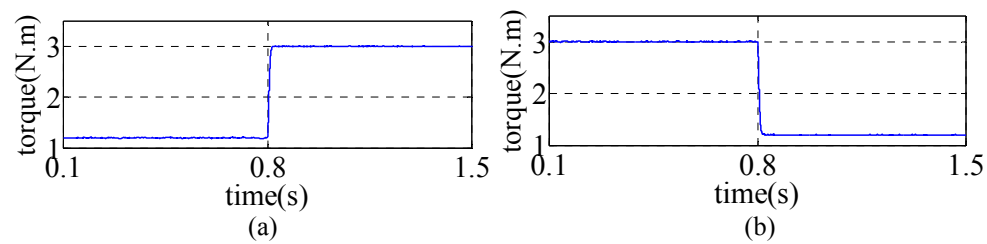


Fig.10. Simulation results of estimated T_L . (c) Estimated load torque in case of sudden load increase from 1.2N.m to 3N.m. (d) Estimated load torque in case of sudden load decrease from 3N.m to 1.2N.m.

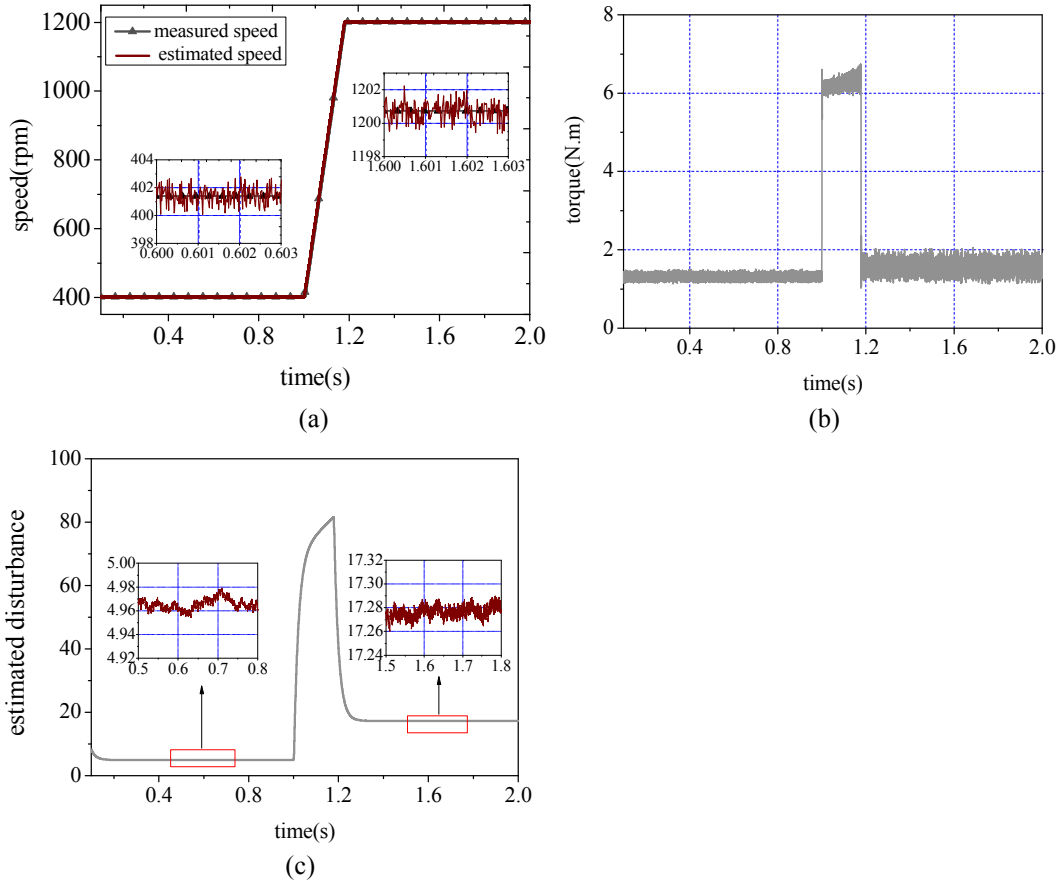
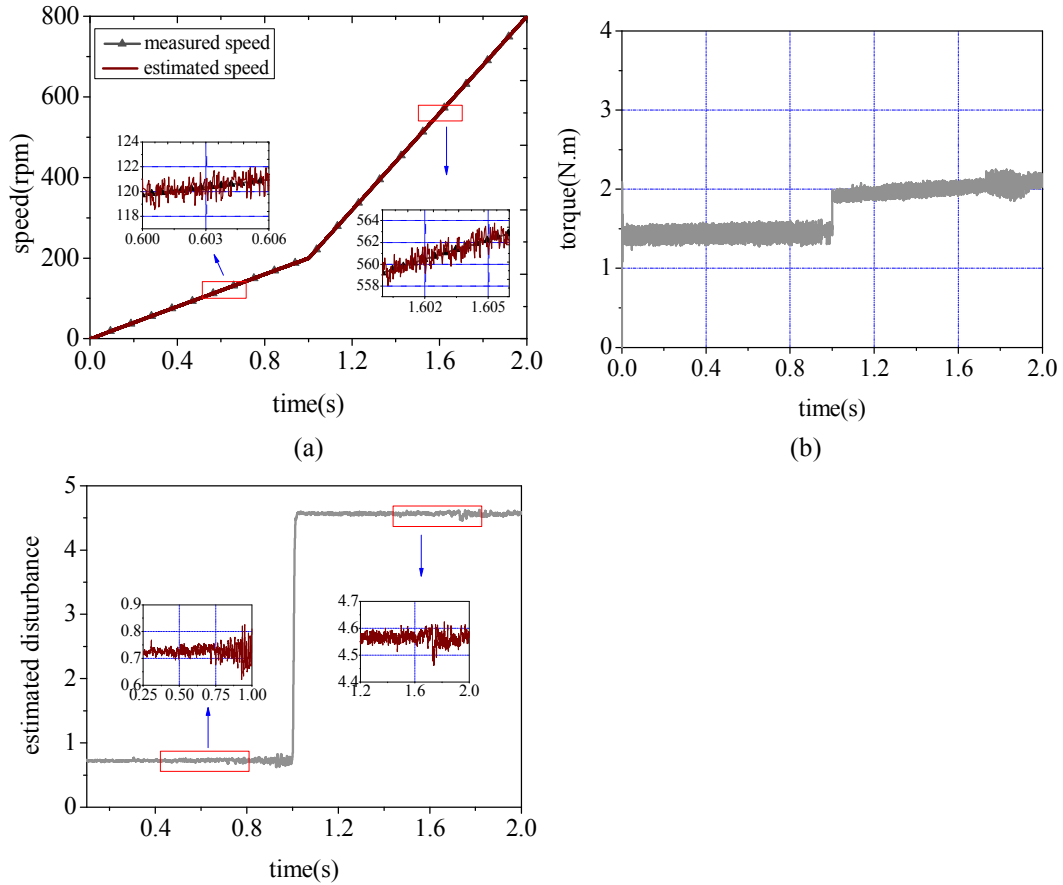


Fig.11. Experimental results when estimating B. (a) Measured speed and estimated speed. (b) Torque. (c) Estimated disturbance in the case of $B_0=50B$.



(c)
Fig. 12. Experimental results when estimating J . (a) Measured speed and estimated speed. (b) Torque. (c) Estimated disturbance in the case of $J_0=10J$.

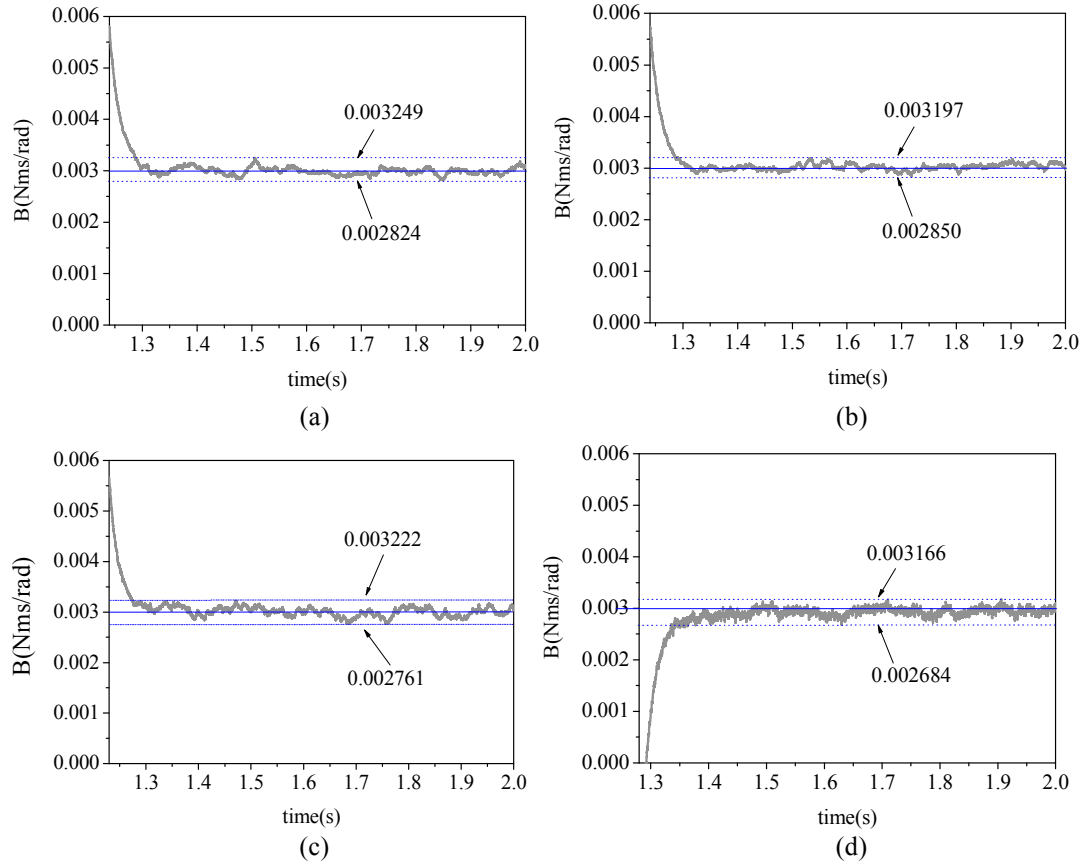


Fig. 13. Experimental results of estimated B . (a) $B_0=0.0001B$. (b) $B_0=0.01B$. (c) $B_0=1.33B$. (d) $B_0=50B$.

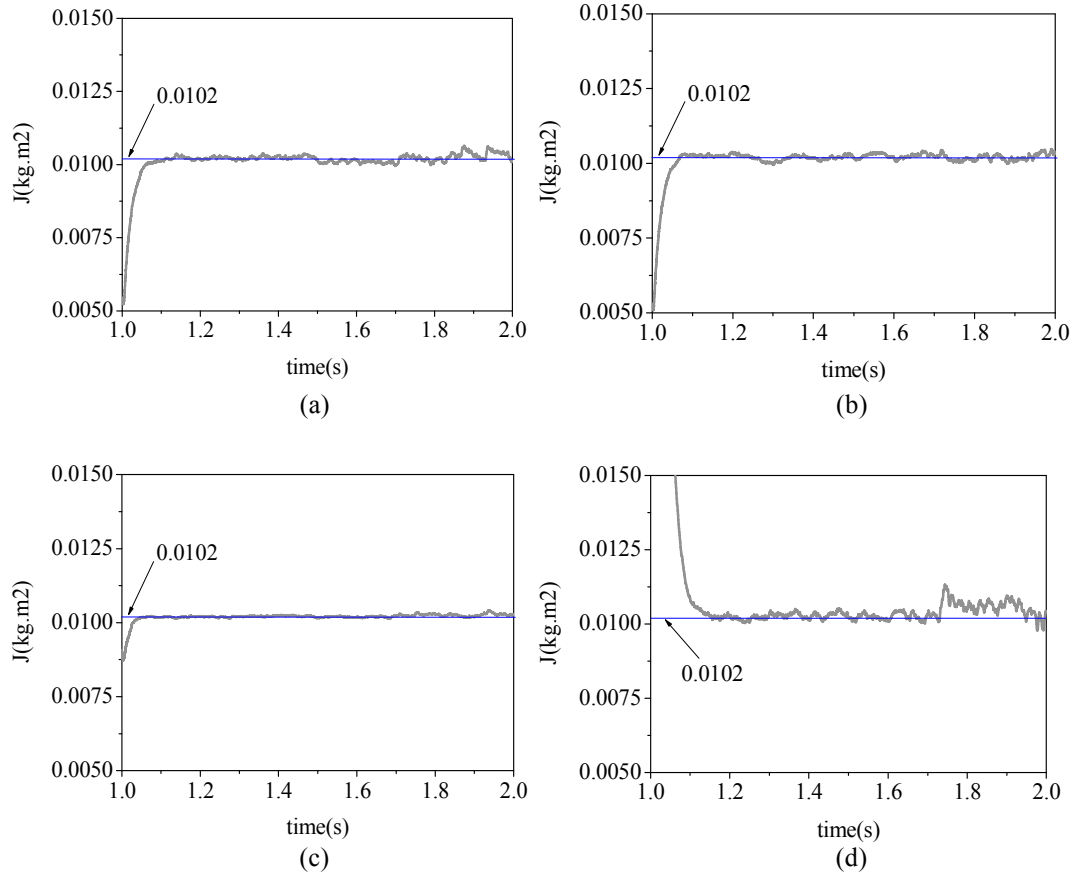


Fig.14. Experimental results of estimated J . (a) $J_0=0.001J$. (b) $J_0=0.01J$. (c) $J_0=2.5J$. (d) $J_0=10J$.

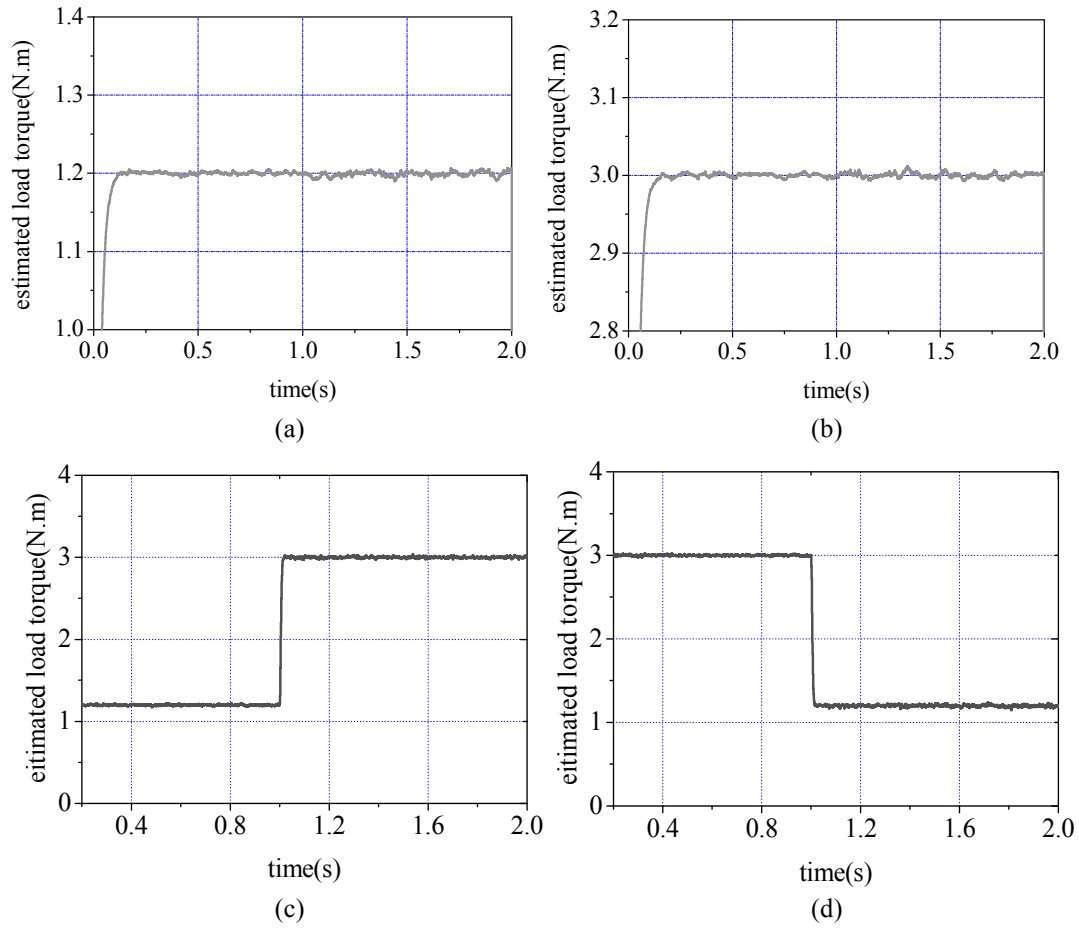
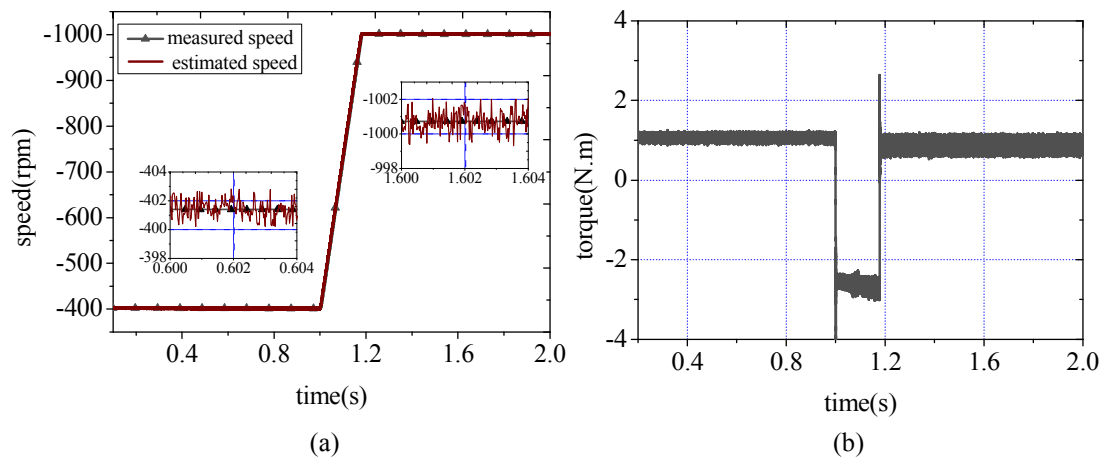


Fig.15. Experimental results of estimated T_L . (a) $T_L=1.2N.m$. (b) $T_L=3N.m$. (c) Estimated load torque in case of sudden load increase from 1.2N.m to 3N.m. (d) Estimated load torque in case of sudden load decrease from 3N.m to 1.2N.m.



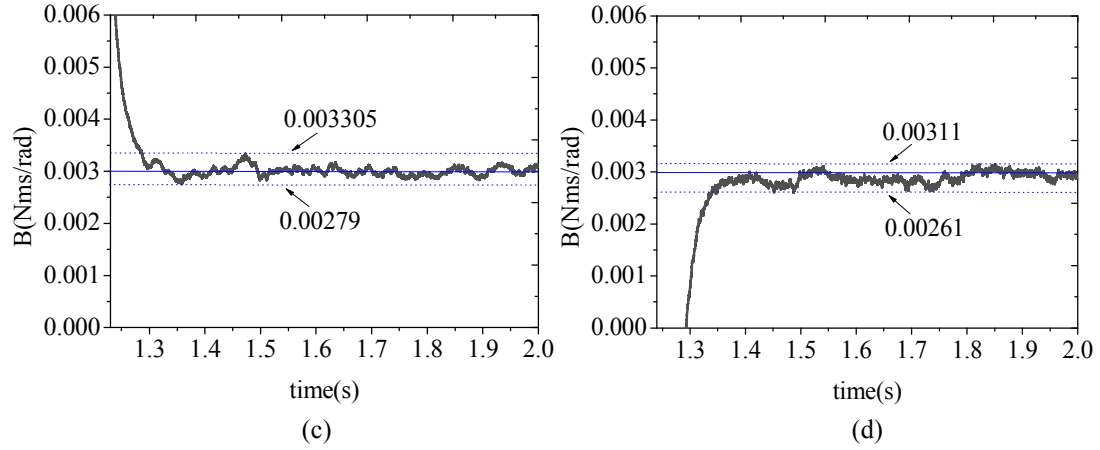


Fig.16. Experimental results of estimated B under speed reversals. (a) Measured speed and estimated speed. (b) Torque. (c) $B_0=0.0001B$. (d) $B_0=50B$.

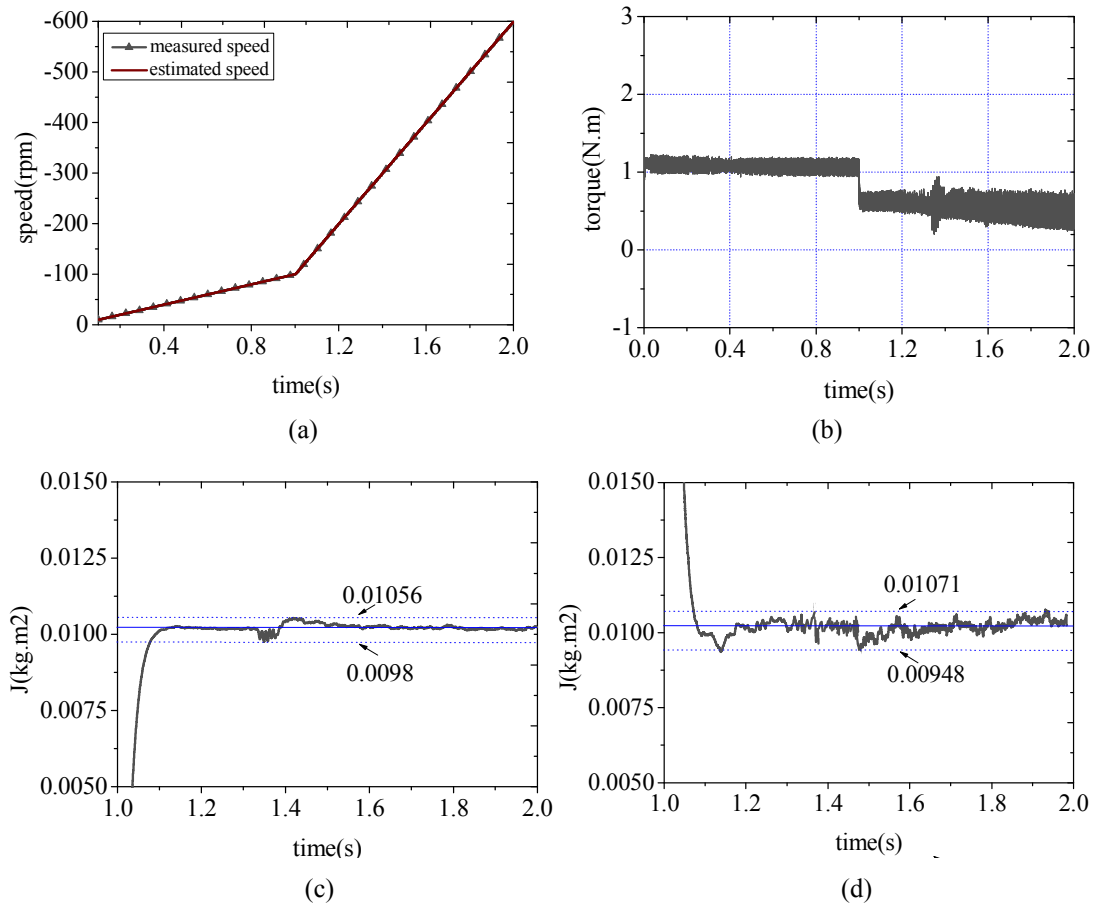


Fig.17. Experimental results of estimated J under acceleration reversals. (a) Measured speed and estimated speed. (b) Torque. (c) $J_0=0.001J$. (d) $J_0=10J$.

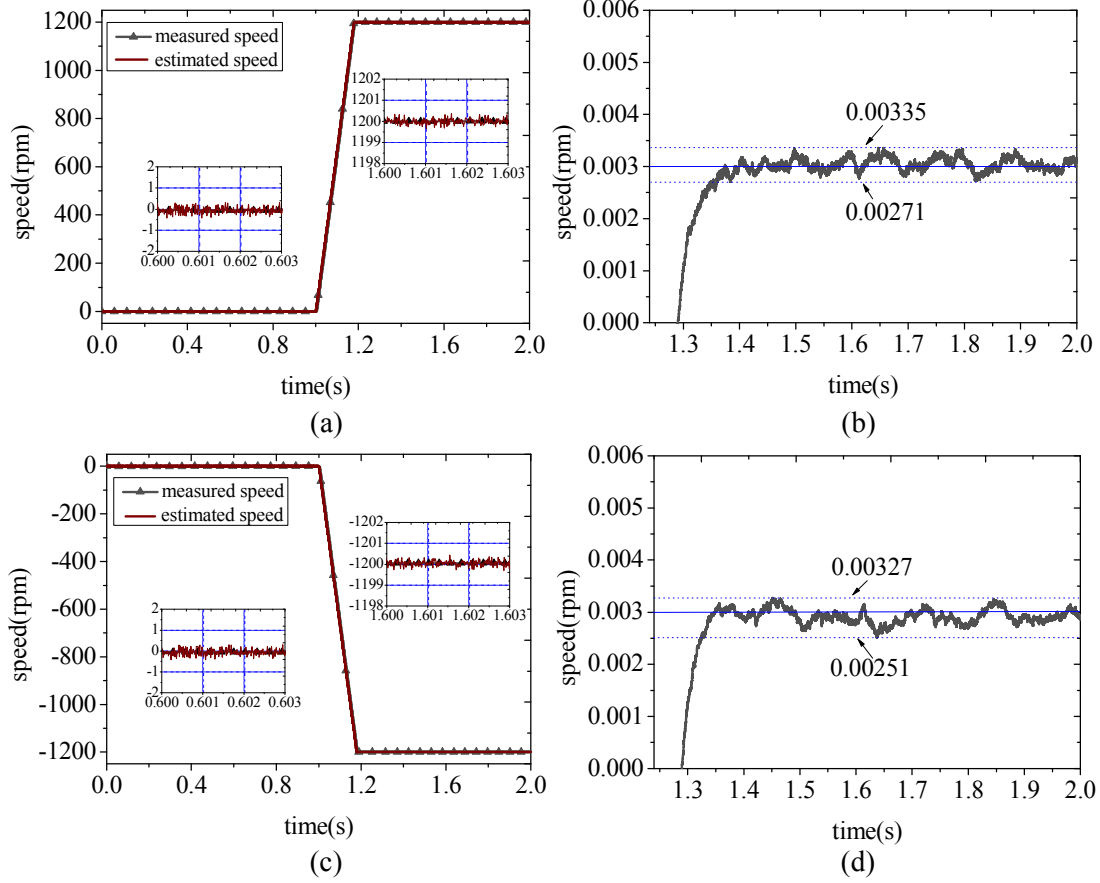
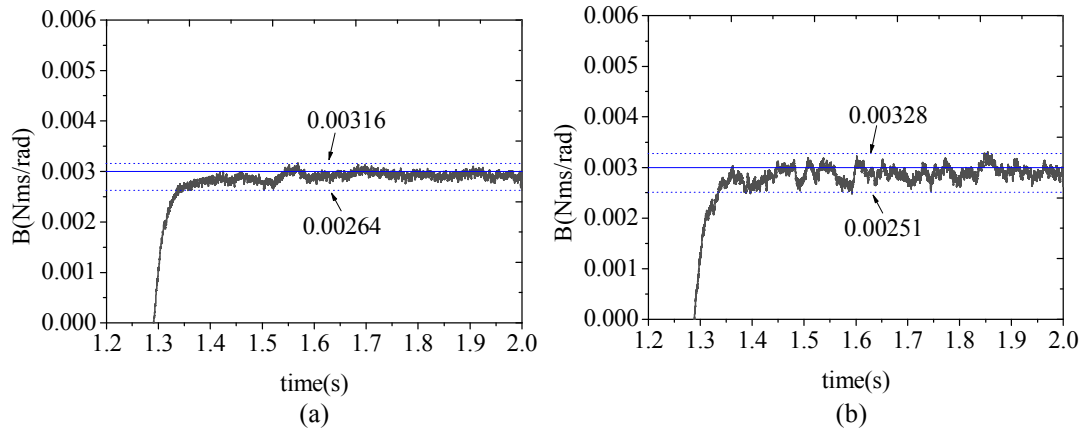


Fig.18. Experimental results of estimated B at zero speed with $B_0=50B$. (a) Measured speed and estimated speed when speed change from 0rpm to 1200rpm. (b) Estimated B when speed change from 0rpm to 1200rpm. (c) Measured speed and estimated speed when speed change from 0rpm to -1200rpm. (d) Estimated B when speed change from 0rpm to -1200rpm.



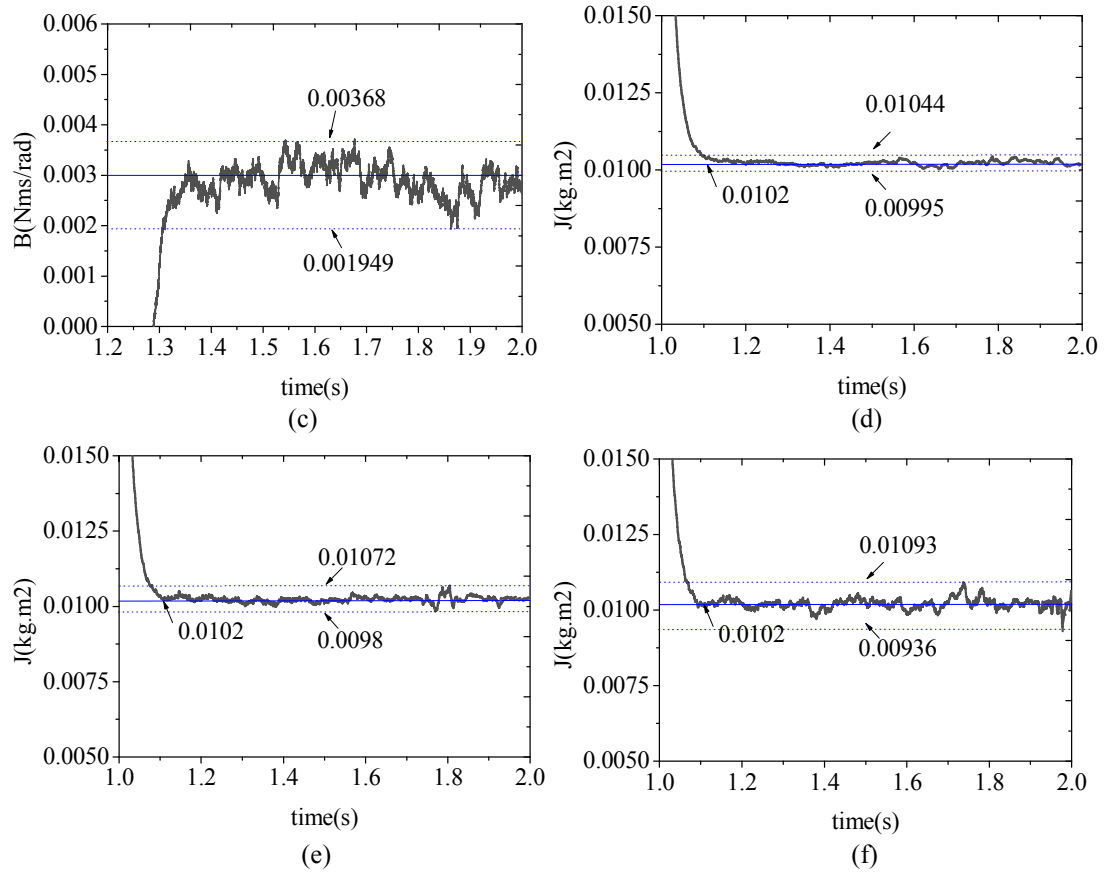


Fig. 19 Experiment results of parameter estimation with three different cut-off frequency of equivalent low-pass filter. (a) Estimated B with $m=2$, $B_0=60B$. (b) Estimated B with $m=5$, $B_0=60B$. (c) Estimated B with $m=10$, $B_0=60B$. (d) Estimated J with $m=2$, $J_0=5J$. (e) Estimated J with $m=5$, $J_0=5J$. (f) Estimated J with $m=10$, $J_0=5J$.

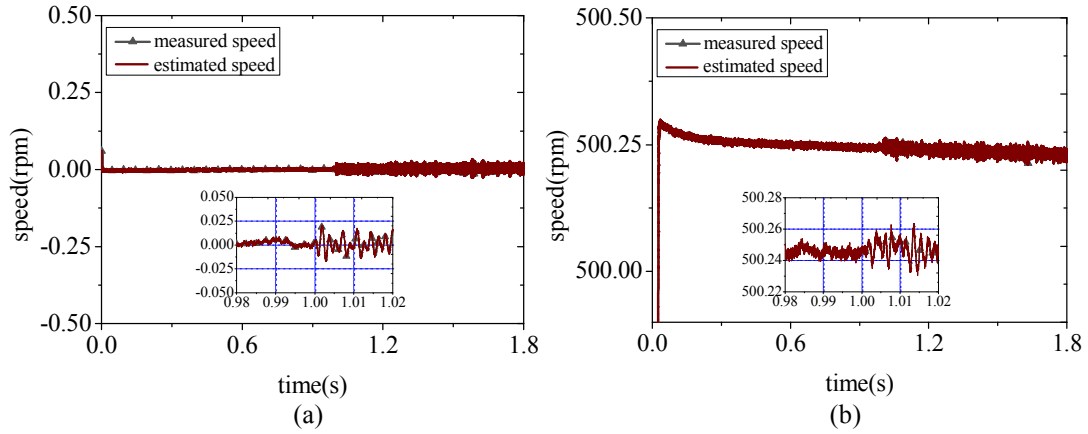


Fig. 20 Experiment results of estimated speed and measured speed when speed feedback signal switching from measured speed to estimated speed at 1 second. (a) Speed response under zero speed. (b) Speed response under 500 rpm.

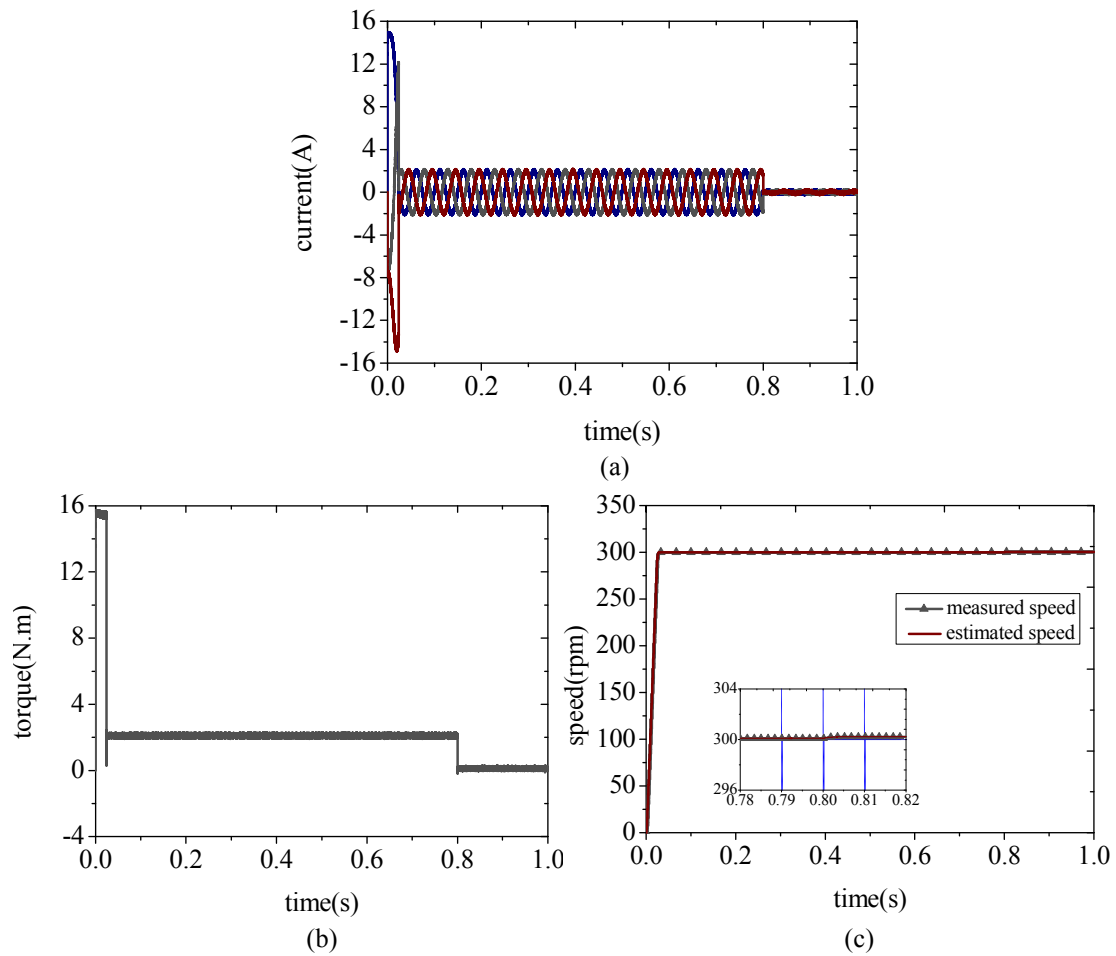


Fig. 21 Experimental results of utilizing estimated parameter in speed controller. (a) Three phase currents. (b) Electromagnetic torque. (c) Speed response.

TABLE I
Parameters of PMSM Control System

d- and q-axes inductances	$L_d = L_q = 9\text{mH}$
Stator phase resistance	$R = 2.6\Omega$
Viscous friction coefficient	$B = 0.003\text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$
Number of pole pairs	$P = 4$
Rotational inertia	$J = 0.0102\text{kg}\cdot\text{m}^2$
Flux linkage of permanent magnets	$\psi_a = 0.175\text{Wb}$

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